

Analyses of ping-pong handovers in real 4G telecommunication networks

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ABSTRACT

The last decade has been characterized by a rapid increase in the usage of mobile communications. One of the main aspects of mobile communications is mobility. This means that mobile phones have to switch between base station cells in order to support the uninterrupted usage of all available services within the area of the mobile network coverage. The process of switching user devices between base station cells is called a handover. Accordingly, base stations are optimized to serve mobile phones with certain moving velocities based on an area and user characteristics to better handle handovers. However, issues appear when a mobile phone starts switching rapidly between two different cells in a short period of time, referred to as ping-pong handover. These occurrences cause negative effects in terms of increased telecom operators signaling overhead and user dissatisfaction. The 3rd Generation Partnership Project (3GPP) defines a metric called ping-pong rate in order to analyze what is the most adequate base station configuration concerning different user velocities and how well base stations handle handovers. The ping-pong rate is defined as a share of ping-pong handovers in total handovers, without taking into consideration the conditions under which they occur. In this paper, the conditions under which ping-pong handovers occur are analyzed by using live, non-simulated data. Findings show that the handovers under different conditions have different probabilities to become ping-pong handovers. Furthermore, these findings are utilized for defining a new empiric metric used for evaluating the handover handling of pre-configured base stations. Consequently, the novel empiric metric provides better insights into the base station handling of handovers than the ping-pong rate defined by 3GPP.

1. Introduction

Over the past few decades mobile communications, along with data communications, have been growing rapidly [1]. Along with the increase in the number of 4G subscriptions from 9 million in 2011 to 4.7 billion in 2021, Ericsson reported that mobile networks carry 300 times more mobile data traffic than in 2011 [2]. The main features of the 4th generation (4G) of mobile networks are high speed and capacity with low cost per transferred bit, better scheduling and call admission control techniques, seamless switching along with a variety of Quality-of-Service (QoS) driven services, and support for interactive multimedia, voice streaming, video, Internet, and other broadband services [3]. Among those, seamless switching represents one of the most important aspects of mobile communications, which is referred to as mobility [4].

1.1. Mobility aspect in 4G

The mobility aspect denotes that the user equipment, which in this case is a 4G-enabled device, can move its position freely within

the coverage of 4G network while remaining connected. During this movement, the user equipment can use all available services, including data traffic and phone calls, with an ensured and appropriate level of QoS [5]. To fulfill the mobility aspect, a handover mechanism in 4G networks has been developed for seamless switching of the user equipment between 4G base stations, or more specifically between directed antennas of the base stations, referred to as cells [6]. Handovers ought to be fast and seamless with minimal delay to ensure connection stability and QoS at all times (especially when there is an active phone call or data session).

Since 4G only supports hard handovers (break before make, i.e., the ongoing connection is broken before a new connection is established), there is a delay in switching user equipment from one base station to another, which must be minimized to ensure connectivity along with the required QoS [7]. In addition, 4G has requirements to support mobility with high performance concerning user equipment velocity of up to 120 km/h [7]. This does not mean that 4G does not support user equipment velocities above 120 km/h. Rather, it means that the performance of 4G networks is limited, as it is reported that 4G network

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in high-speed trains can handle user equipment velocities of up to 350–500 km/h [5].

1.2. Ping-pong handover problem definition

Handovers are negatively affected by the occurrence of handover failures and ping-pong handovers [8]. On the one hand, handover failures represent unsuccessful handovers [9] that cause signal loss along with a phone call or an active data session interruption. On the other hand, ping-pong handovers are situations when user equipment rapidly switches between two cells within a short period of time, creating the unnecessary signaling overhead at base stations and degrading QoS for other user equipment [8]. In addition to the user dissatisfaction due to poor network performance [10], ping-pong handovers cause significant resource problems for telecom operators, as they waste limited resources by reserving two connections with two base station cells simultaneously for a certain period of time [11].

The main reasons for ping-pong handovers are user equipment movements between base stations and high variations in signal quality between the base stations that compete to provide connection to user equipment [12]. Accordingly, the number of ping-pong handovers has to be minimized [12]. Therefore, the base stations must be configured well to ensure optimal handovers and avoid the negative effects of increased ping-pong handovers. However, there is a trade-off between failed handovers and ping-pong handovers, where reducing the number of ping-pong handovers results in potentially increasing the number of failed handovers and vice versa [13]. Consequently, accurately measuring ping-pong handovers of base stations in the network is an indispensable tool for optimizing configuration and minimizing ping-pong handovers in general.

1.3. Comprehensive approach to managing mobile networks

The current approach for measuring handover performance for a single base station is simply calculating the percentage of ping-pong handovers out of the total number of handovers [9] for that base station, without taking into consideration interdependence between neighboring base stations. This results in a limited view of the network state since handovers happen only in areas where there is more than one base station, i.e., it happens between at least two base stations.

A more comprehensive approach that can observe an entire 4G network as a system of systems is required and it should provide an overview of the entire network state. Moreover, taking each base station and modeling its behavior and interdependence with neighboring base stations using empirical and data-driven approaches is required, i.e., relying heavily on data. Similar approaches are taken when applying Digital Twin concepts [14] on telecommunication networks. Consequently, monitoring handover performance of individual base stations requires a novel metric that utilizes such an approach by observing the base station as part of the entire system and basing its reasoning on live network data.

1.4. Novel metric for expressing ping-pong rate

This paper aims to analyze the conditions under which ping-pong handovers occur and apply those findings to define a novel handover performance metric. Consequently, every base station configuration is observed as a black box system, which means that nothing about the internal configuration is known. The existing 3rd Generation Partnership Project (3GPP) metrics are used to demonstrate the shortcomings of current monitoring approaches. Furthermore, a novel metric that is based on empirical findings acquired from the real network data is presented. To the best of our knowledge, this is one of the few papers that utilize (anonymized) live network data in handover analyses instead of simulated data usage. Results show that the novel handover

performance metric is more accurate than the standard 3GPP metrics since it follows the trend extracted from the live network data.

The rest of the paper is organized into five sections. Section 2 covers the work related to the topic of handover analyses in mobile networks. Handover classification and technical details about the handover procedure are introduced in Section 3. Section 4 gives an overview of the used dataset and methodology employed for performing analyses. Novel handover performance metric and the results of the analyses are presented in Section 5, while Section 6 and Section 7 reveal the discussion and conclusions about the conducted analyses, respectively.

2. Related work

Lots of effort is invested in optimizing handover procedures to avoid unnecessary handovers and establish a good trade-off between ping-pong handovers and failed handovers. In [15], the authors review handover improvement methods along with the performance of different handover configuration parameters. Bae et al. in [16] propose adaptive handover margin algorithm based on cell types. Their algorithm resulted in low number of ping-pong handovers and low handover failures. In [17] Emran et al. provide new solution for managing handovers in LTE network. Their solution reduces delay during handover procedure, thus directly improving the network performance. Feher et al. in [18] propose a method for classifying and reducing unwanted ping-pong handovers. To evaluate the advantages of the proposed algorithm, they use live network measurements. Huaining et al. in [19] introduce a new handover prediction technique to improve handover performance. Simulation results show that when using the technique, the number of handovers and the value of the ping-pong handover rate is reduced. In [20], the authors propose a handover performance indicator for the ping-pong handovers along with a mechanism for tuning handover parameters to reduce unwanted ping-pong handover impact. Their mechanism successfully reduces ping-pong handovers and keeps handover failures at acceptable values. Narvaez and Bonilla in [21] review handover algorithms oriented to massive transport while concluding that, except for standard 3GPP parameters that affect handover occurrence, other parameters such as user equipment velocity, position, and direction are also important for that use case.

Ghanem et al. in [12] investigate ping-pong handover effects in 4G network and propose a ping-pong handover avoidance scheme. Their data consists only of fixed low/mid user equipment velocities (25 and 70 km/h) and to evaluate the proposed schema, TRIAS tool for simulating mobile networks is used. Mittal et al. in [8] estimate overhead in 4G networks during handover and non-handover conditions based on user equipment velocity and application traffic. The simulation environment for testing consists of the Aeroflex simulator, commercial eNodeB, 4G core network simulator, and data traffic generator. According to their experimental results, significant overhead and performance degradation occur during handovers. Tayyab et al. in [22] test the impact of the cell size and user equipment velocity on the handover procedure. Low cell size results show performance degradation, especially when user equipment velocity increases, while large cell size results in increased handover failure for low user equipment velocity.

Dimou et al. in [7] investigate handover performance using handover failure rate and overlay delay metric based on different cell radius, number of user equipment, user equipment velocity, and traffic loads per cell. They use simulation data showing that the handover failure rate increases along with user equipment velocity and system load. Along with these scientific works, Ta-Gang Chiou in [23] defines and presents the patented method for detecting and reducing ping-pong handovers in 4G networks.

To sum up, most of the related work uses simulation data and tries to optimize the existing handover algorithms or propose new ones. Additionally, all of these papers use the 3GPP defined metric for estimating the base station configuration quality. This paper covers 4G data analyses to investigate conditions in which ping-pong handovers occur

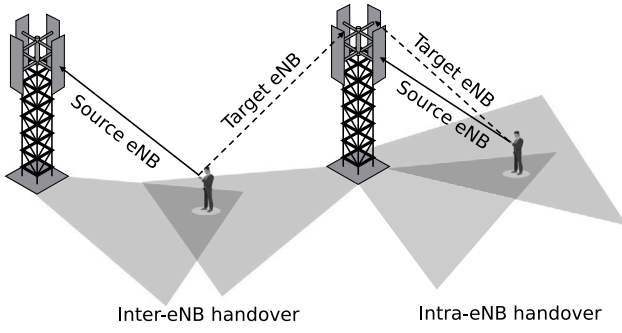


Fig. 1. Handover types.

and presents a novel empiric metric for estimating the configuration quality of base stations. To the best of our knowledge, this is the first paper that defines a new metric for estimating the base station configuration quality based on ping pong handovers, along with 3GPP defined metric. Furthermore, in comparison with the existing related works which use simulated data, the most important difference is that the analyses presented in this work are completely based on real data obtained from a live 4G network.

3. 4G handover concepts

In 4G networks, handover (HO) is an essential process to ensure basic network functionality, i.e., mobility. Hence, an appropriate configuration of a base station (eNB) to perform HO when it is necessary is important. This comes with its own set of challenges. On the one hand, frequent HOs cause unnecessary signaling overhead on an eNB. On the other hand, infrequent HOs can cause a weak signal level on user equipment (UE) or even complete loss of connectivity. Consequently, properly configuring an eNB for performing appropriate HOs is important for achieving good and stable cellular network operation.

3.1. Handover classification

There are two types of HO processes: the intra-eNB handover and the inter-eNB handover [8] (Fig. 1). The intra-eNB handover represents a handover between two cells of the same eNB, while the inter-eNB handover is a handover between two cells of two different eNBs. In this paper, all types of HOs are observed as the same group.

The 3GPP standard defines the ping-pong handover (PP) as a handover where a UE switches from a cell c_i to a cell c_j and then connects back to the cell c_i within a time of stay (TS) which lasts less than the minimum time of stay (MTS) equal to 1 s [9]. Therefore, the TS represents a time period during which the UE is connected to one cell [9] and if the duration of this period is less than MTS, the PP has occurred. PPs appearance is possible in all types of HOs.

3.2. Handover process

According to the 4G standard [6], the hysteresis margin (HOM) and the time-to-trigger (TTT) are two crucial parameters that control HO occurrences, as depicted in Fig. 2, which illustrates changes in the Reference Signal Received Power (RSRP) in time for a UE while within the coverage of two eNBs. The HOM defines a constant value (in dBm) that represents a threshold for the difference between levels of two eNB signals received by the UE [13]. The TTT defines a time period (in seconds) during which the HOM condition must be satisfied, i.e., it defines the time period during which the difference between signal levels (RSRPs) of the eNBs at the position of the UE must be higher than the threshold. Therefore, the HO will be triggered if the HOM level remains fulfilled for the TTT period [13]. HOM and TTT parameters are

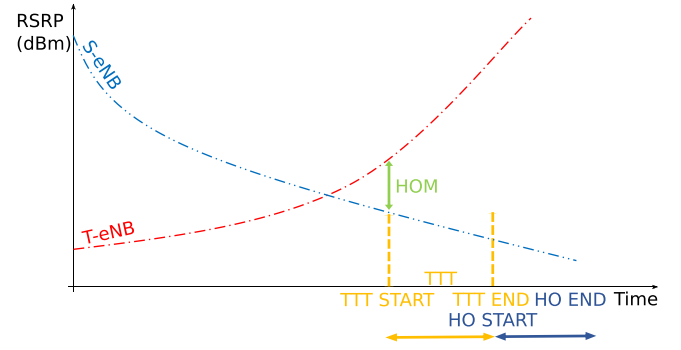


Fig. 2. eNB signal level changes during the handover process.

Table 1

Dataset header fields.

Data field	Description	Example
IMSI	Anonymized sim card id	019R1cDccWUfb4ejVzFvYm
BS name	Connected eNB	Base station 1
CellID	Connected CellID	123456
Geometry	UE Location	POINT (17.95618 59.40470)
Timestamp	Connection timestamp	2020-01-01T10:00:00.000

Table 2

Mediterranean area statistics.

Feature	Notation	Value
Total number of UEs	$ U $	276 513
Total number of Cells	$ C $	2 856
Total number of eNBs	$ B $	426
The average number of cells per eNB	$ C^b $	6.704225

configured based on area characteristics and traffic requirements using expert knowledge.

In practice, the HO process among two eNB cells goes as follows: a UE connected to an eNB (called source eNB or S-eNB) cell (c_i) scans all other available eNBs in the area and sends a report to the S-eNB. In a situation when a signal from another eNB cell (c_j) becomes stronger than the signal of the S-eNB cell (c_i) as hysteresis margin (HOM) defines it (Fig. 2) and it remains stronger for the time-to-trigger (TTT) period, the S-eNB decides to handover the UE to the target eNB (T-eNB). If all subsequent steps related to the signaling information transfer after the HO decision are successfully completed, the UE is handed over to the T-eNB, which becomes a new S-eNB [13].

4. Dataset and methodology

The telecom data used in this study represent a dataset from a real telecommunication network. The recorded data contain information regarding which (anonymized) UEs were connected, to which eNB cells (antennas), at what time, and at what location. Table 1 shows all dataset fields, along with their descriptions and examples.

The dataset comprises the above information collected over 13 days, in a region of approximately 14 882 km² in a Mediterranean area. The total number of records is around two hundred eighty-seven million. Table 2 summarizes the basic statistics of the dataset.

The following Expressions (1), (2) and (3) introduce different types of notation. Thus B represents a set of all eNBs allocated in the analyzed area, C^b represents all the cells of each eNB b and U represents a set of all UEs. Finally, the S^u represents an ordered set of data records r_i for the UE u , sorted by timestamps. According to Expression (4), the record r_i is composed of a set of triples which includes information about the cell of the eNB b (c_i^b) to which the UE u is connected during the i -th

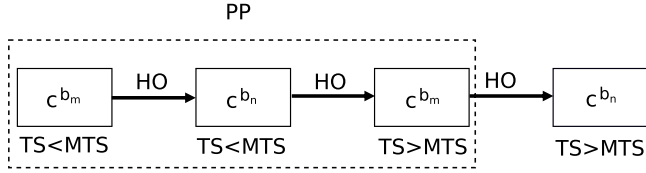
Algorithm 1: Counting handovers**Data:** All available records**Result:** Number of handovers per eNBs

```

1  $\forall u \in \mathbb{U}$ 
2    $\forall (r_i, r_{i+1}) \in \mathbb{S}^u$ 
3     if  $(c_i^{b_m} \neq c_{i+1}^{b_n})$ 
4        $|HO^{b_m}| = |HO^{b_m}| + 1$ 

```

for every UE
 # for pair of records of single UE
 # if the source cell and the target cell are different
 # increase number of HOs for b_m

**Fig. 3.** Counting handovers (HOs) and ping-pong handovers (PPs).

record and the geographic coordinates of the UE (s_i) in the timestamp (t_i) when the i -th record is recorded.

$$\mathbb{B} = \{b : b \text{ is an eNB}\} \quad (1)$$

$$\mathbb{C}^b = \{c^b : c^b \text{ is a cell of eNB } b \in \mathbb{B}\} \quad (2)$$

$$\mathbb{U} = \{u : u \text{ is a UE}\} \quad (3)$$

$$\mathbb{S}^u = (r_i = (c_i^b, s_i, t_i) : r_i \text{ is } i\text{th record of user } u \in \mathbb{U}, \\ c_i^b \in \mathbb{C}^b, b \in \mathbb{B}, t_i < t_{i+1} \forall i, \\ s_i \text{ are coordinates, } t_i \text{ is a timestamp}) \quad (4)$$

4.1. Detecting handovers and ping-pong handovers

It is important for this analysis that HO events are correctly and accurately detected. Fig. 3 depicts how to count HOs and PPs for a single UE. In this example, there are three HOs and one PP. Note that HO sequence $c^{b_n} \rightarrow c^{b_m} \rightarrow c^{b_n}$ in Fig. 3 is not a PP, as it does not satisfy the condition that the time of stay (TS) must be less than the minimum time of stay (MTS) when a UE is connected to c^{b_m} .

To be able to count HOs and PPs, the transformation must be applied to the raw data. Expression (5) depicts the HO definition using the proposed notation.

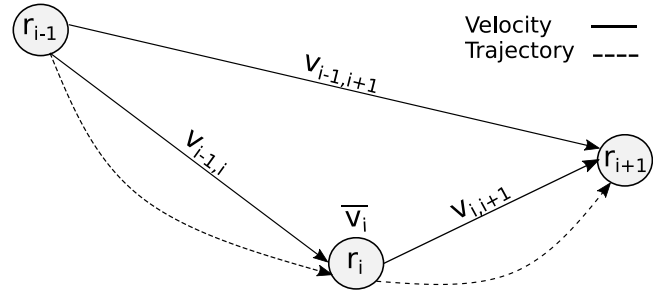
$$(r_i, r_{i+1}) \text{ is HO} \iff c_i^{b_m} \neq c_{i+1}^{b_n} : r_i, r_{i+1} \in \mathbb{S}^u, u \in \mathbb{U} \quad (5)$$

According to Expression (5), two consecutive data records (r_i, r_{i+1}) represent HO only if those records belong to different eNB cells ($c_i^{b_m} \neq c_{i+1}^{b_n}$). Algorithm 1 is used to count HOs in all available records, where $|HO^{b_m}|$ represents a number of HOs for eNB b_m . According to Algorithm 1, for every UE and every pair of records of a single UE, and if the source cell and the target cell are different, the number of HOs ($|HO^{b_m}|$) will be increased for the eNB b_m . When counting the HO events, it is important to remove all the non-HO events, as well as to ensure that all the detected HOs are related to the same UE. Finally, HO defined as $c^{b_m} \rightarrow c^{b_n}$ is counted as a HO for the eNB b_m .

According to Expression (6), the PPs counting in a formal manner is defined as:

$$(r_i, r_{i+1}, r_{i+2}) \text{ is PP} \iff (r_i, r_{i+1}) \text{ is HO} \wedge \\ (r_{i+1}, r_{i+2}) \text{ is HO} \wedge \\ t_{i+2} - t_{i+1} < 1s \wedge \\ c_i^{b_m} = c_{i+2}^{b_m} : r_i, r_{i+1}, r_{i+2} \in \mathbb{S}^u, u \in \mathbb{U} \quad (6)$$

Counting of the PPs is performed through seven steps of Algorithm 2. As was the case with the HOs counting performed by Algorithm 1, when

**Fig. 4.** All locations/velocities used for UE velocity approximation in record r_i .

counting the PPs it is equally important to remove all non-PP events and ensure that the rest of them belong to the same UE. Counting the PPs ends in a way that a sequence of HO events $c^{b_m} \rightarrow c^{b_n} \rightarrow c^{b_m}$ in Fig. 3 is counted as the PP for the eNB b_n . The notation $|PP^{b_n}|$ defined in the seventh step of Algorithm 2 represents the number of PPs for eNB b_n .

4.2. Approximation of UE velocity

Each eNB has a different configuration, which is mostly determined by the surrounding environment and serving UEs. One of the important factors that impact an eNB configuration is the UE velocity. It represents the expected average movement velocity of the UEs covered by a specific eNB. This factor is commonly estimated and used as a guideline for the eNBs configuration. However, the UE velocity is not available inside the obtained mobile operator dataset. Thus, an approximation is performed based on the available data.

A general formula for expressing UE velocity ($v_{i,j}$) based on two records, r_i and r_j , is given in Eq. (7):

$$v_{i,j} = \frac{|s_j - s_i|}{t_j - t_i} : r_i, r_j \in \mathbb{S}^u, u \in \mathbb{U} \quad (7)$$

where s_j and s_i represent location coordinates of i -th and j -th UE records and t_j and t_i represent the timestamps of the same records.

Nevertheless, only two records for the UE velocity approximation cannot be used, since information about UE locations obtained from records r_i and r_j is only reliable to some degree. Therefore, to reduce the unwanted uncertainty related to the accuracy of the UE location obtained from only two records, a new formula is defined in Eq. (8):

$$\bar{v}_i = \frac{v_{i-1,i} + v_{i,i+1} + v_{i-1,i+1}}{3} : r_{i-1}, r_i, r_{i+1} \in \mathbb{S}^u, u \in \mathbb{U} \quad (8)$$

It offers a more accurate approximation of the UE velocity $\bar{v}_i$ for $r_i \in \mathbb{S}^u$. Fig. 4 illustrates an example of UE velocity calculation. In Fig. 4, all locations/velocities that are used for the calculation are graphically displayed. According to Fig. 4, UE velocity calculation is performed by taking into account velocities of the UE using three UE records characterizing the UE state in i , $i-1$ and $i+1$ timestamp.

Since the UE velocity is a continuous variable, for the analyses, it is discretized into several intervals. Furthermore, the reason therefore is to categorize UE velocities into different profiles and to get insights into eNBs behavior in different UE velocity profiles. The UE velocity

Algorithm 2: Counting ping-pong handovers**Data:** All available records**Result:** Number of ping-pong handovers per eNBs

```

1  $\forall u \in \mathbb{U}$ 
2  $\forall (r_i, r_{i+1}, r_{i+2}) \in \mathbb{S}^u$ 
3   if (  $c_i^{b_m} \neq c_{i+1}^{b_n}$ 
4      $\wedge c_{i+1}^{b_n} \neq c_{i+2}^{b_m}$ 
5      $\wedge c_i^{b_m} = c_{i+2}^{b_m}$ 
6      $\wedge t_{i+2} - t_{i+1} < 1s$  )
7      $|PP^{b_n}| = |PP^{b_n}| + 1$ 

```

for every UE
for triple of records of single UE
if first and second cells are different
and second and third are different
and first and third are the same
and TS in second is less than MTS
increase number of PPs for the second eNB

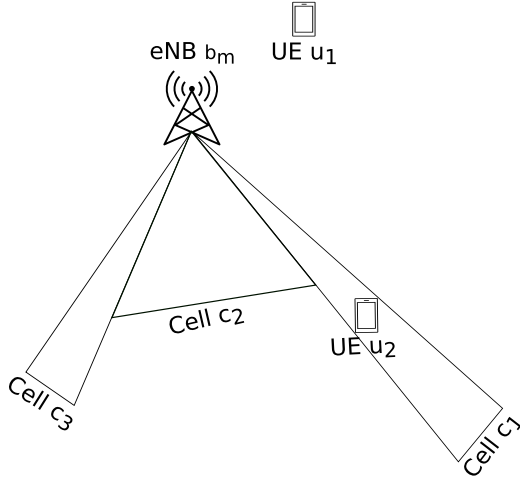


Fig. 5. Physical distance approximation problem.

values are divided into intervals in the form of $[min(v), max(v))$, unit is km/h, and the upper bound in a final interval is included in the interval. Expression (9) defines a set that represents all used UE velocity intervals $\mathbb{V}$ in a way that:

$$\mathbb{V} = \{[0, 10), [10, 20), \dots, [100, 110), [110, 120]\}$$

$$f : [0, 120] \rightarrow \mathbb{V} \quad (9)$$

$$f(\bar{v}_i) = V_j \iff \bar{v}_i \in V_j, V_j \in \mathbb{V}$$

According to Expression (9), starting UE velocity is 0 km/h with a step of 10 km/h for each interval, while the maximum UE velocity is 120 km/h. Due to the requirement for mobility support in 4G networks, which need to ensure high performance regarding UE velocity of up to 120 km/h, the maximum UE velocity is set to 120 km/h [7]. All UE velocities above 120 km/h are discarded.

4.3. Approximating eNB coverage

Apart from the calculated UE velocity, a metric is required for observing conditions in which HOs occur. Such a metric must reflect the relationship between a UE and the connected eNB. Physical distance, as the first intuitive metric, is problematic due to eNB cells being commonly installed using directed antennas. Consequently, that means that the UE can be physically near eNB, but not within its coverage since all eNB cells might be directed elsewhere. Fig. 5 depicts a physical distance problem for two UEs, and according to the example presented in the figure, the UE u_1 (that is not even within the coverage of eNB) is in terms of physical distance closer than UE u_2 . Furthermore, HOs occur in places where at least two eNBs are competing for serving UEs in that area, and the physical distance cannot give any indication of competing eNBs.

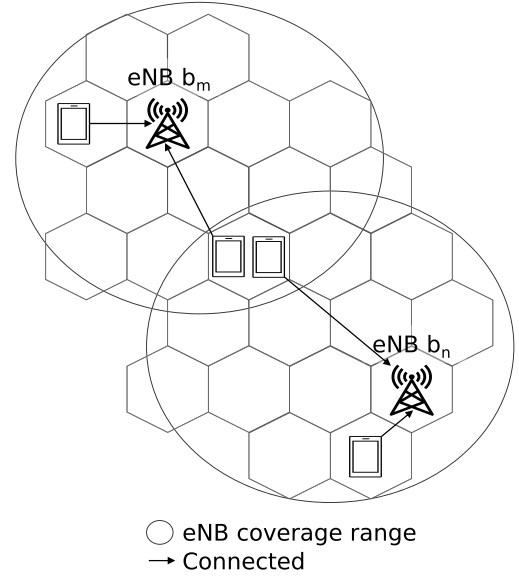


Fig. 6. Discrete global grid ISEA3H area units (hexagons) and eNBs spatial relation.

Due to the previously described physical distance problem, in these analyses, the eNB coverage metric is defined as a metric that represents a percentage of UEs that are served by the eNB within a specific area a . eNB coverage provides a good indication of spots/locations where multiple eNBs are competing for domination in establishing the connection with UEs which are located within the coverage area of that eNB. Consequently, such a metric should describe relationship aspects between the UEs and corresponding eNBs, with regard to their connectivity. Accordingly, an area around eNBs is divided into smaller units (hexagons) using a discrete global grid ISEA3H [24] depicted in Fig. 6. Each hexagon that forms the global grid has a diameter of 907 m and covers an area of approximately 0.53 km². The set of area units can be defined as:

$$\mathbb{A} = \{a : a \text{ an area unit (hexagon)}\} \quad (10)$$

The area units are used for tagging individual UE records r_i with the eNB coverage metric.

Fig. 6 shows the eNBs positions and their coverage range inside an area divided into multiple hexagons. On the one hand, UEs that pass through hexagons where the coverage range overlaps in Fig. 6 can connect to b_m or b_n . Hence, these two eNBs will compete for domination in connecting UEs located in the overlapping range area. On the other hand, UEs in a hexagon where eNB b_n is located will always be connected to eNB b_n , since there are no other eNBs to compete for domination over UEs.

For calculating coverage of every eNB ($\mathbb{B}$ from Expression (1)) in each area unit ($\mathbb{A}$ from Expression (10)), only certain subsets of data are used, namely the subsets $\mathbb{S}_a^u$ which contain the sequential records

of a UE u that is located within the area unit a . The set containing all the UE u trajectory segments $\widehat{S}_a^u$ can be defined as

$$\widehat{S}_a^u = \{S_a^u : S_a^u \subseteq S^u, u \in \mathbb{U}, S_a^u = \{r_i = (c_i^b, s_i, t_i) : i = \min, \dots, \max, s_i \in a \forall i \wedge s_{\min-1} \notin a \wedge s_{\max+1} \notin a\}\} \quad (11)$$

The set $\widehat{S}_a^u$ is composed of subsets representing trajectory segments which are defined from the moment when the UE has entered the area unit a until it left it. The total number of distinct UEs within that area unit a is counted as $|\mathbb{U}_a|$. It represents a number of UEs located within the area unit a , where $|\mathbb{U}_a^b| \subseteq |\mathbb{U}_a|$ is for UEs that are also connected to eNB b . Moreover, a total travel time T_a^u for each UE within that area unit a is calculated using Eq. (12).

$$T_a^u = \sum_{S_a^u} (t_{\max} - t_{\min}), r_{\max}, r_{\min} \in S_a^u \quad (12)$$

The total moving (travel) time represents the time a UE has spent traveling within an area unit a . It is calculated as a sum of all periods that UE u spent in the area unit a defined in $\widehat{S}_a^u$. The time spent in the area unit a is calculated using Eq. (12) as the difference between timestamps indicating leave ($t_{\max}$) and entrance ($t_{\min}$) of the UE u from the area unit a detected in records $r_{\max}$ and $r_{\min}$, respectively. Finally, the eNB coverage d_a^b for eNB b within an area unit a is calculated using Eq. (13):

$$d_a^b = \sqrt{\frac{|\mathbb{U}_a^b|}{|\mathbb{U}_a|} \times \frac{\sum_{u \in \mathbb{U}_a^b} T_a^u}{\sum_{u \in \mathbb{U}_a} T_a^u}} \quad (13)$$

Therefore, the numerator of Eq. (13) represents all UEs connected to the eNB b in the area unit a multiplied by the total travel time of UEs that are connected to the eNB b in the area unit a , while the denominator represents all UEs in the area unit a that are connected to all eNBs covering that area, multiplied by the total travel time of all UEs in the area unit a . Finally, to get normalized values, eNB coverage from Eq. (13) is calculated as a square root of two multiplied ratios explained above.

Possible values of the eNB coverage metric are in a range from 0 to 100 percent, where 100 percent means that the eNB has total domination in that specific area unit, i.e., there is no other eNB to compete for establishing a connection with UEs. As the value of the eNB coverage decreases, this domination becomes lower, i.e., there is a lower percentage of UEs connected to that eNB in that specific area unit. Therefore, 100 percent of the eNB coverage metric means that all UEs are connected to the specific eNB in that area, while 0 percent means that no UEs are connected to the specific eNB in that area.

Since eNB coverage metric can take any continuous value between 0 and 100 percent, for analyses to get an insight into the importance of eNB coverage values, eNB coverage metrics are grouped into multiple intervals as follows:

$$\begin{aligned} \mathbb{D} &= \{[00, 10], [10, 20], \dots, [80, 90], [90, 100]\} \\ f : [0, 100] &\rightarrow \mathbb{D} \\ f(d) = D_j &\iff d \in D_j, D_j \in \mathbb{D} \end{aligned} \quad (14)$$

The reason for grouping values of eNB coverage into ranges is to get insights into eNBs behavior for different eNB coverage profiles. The grouping is performed in the form of $[\min(d), \max(d))$, where the upper bound in a final interval is included in the interval. Expression (14) introduces a set definition that represents all defined eNB coverage intervals. The lowest eNB coverage metric is in an interval starting with 0 percent and having a step of 10 percent, while the maximum is 100 percent.

5. Ping-pong rate calculation

To analyze how well a pre-configured eNB handles HOs, the 3GPP defines a metric called ping-pong rate (PPR) [9]. In this section, the limitations of PPR as a metric are examined and a novel and improved

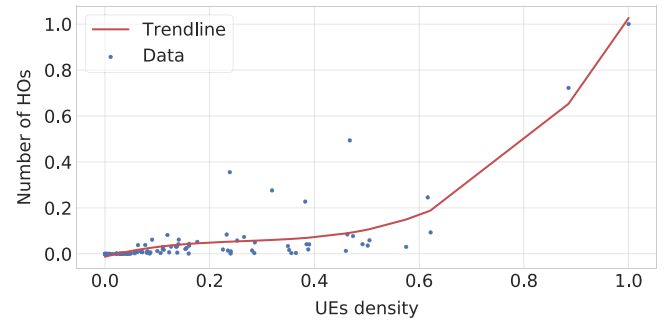


Fig. 7. Number of HOs by UEs density.

empirical metric $\widehat{PPR}$ for expressing ping-pong rate is proposed. The novel ping-pong rate metric ($\widehat{PPR}$) is based on a comprehensive view of the network, hence it utilizes both UE velocity and eNB coverage metric defined in Sections 4.2 and 4.3, respectively.

5.1. Area profiling and eNB classification

Before any evaluation, spatial analysis of the dataset mentioned in Section 4 is performed. Unit area hexagons spreading over approximately 14 882 km² are used to divide the area into smaller area units and inspect how UEs are distributed. A number of distinct UEs in one hexagon in this case is interpreted as the UEs density of that hexagon. Additionally, a number of HOs by UEs density is used to identify areas of interest and select eNBs for detailed analyses. Fig. 7 shows the normalized number of HOs by normalized UEs density values, as well as the trend line with R^2 score of 0.72. According to Fig. 7, the number of the HOs increases with an increase in the UEs density.

According to the European Telecommunications Standards Institute, [25] and the Next Generation Mobile Networks Alliance [26], cellular network coverage areas based on UEs density (represented by a number of UEs per square kilometer) are classified as Urban, Suburban and Rural coverage areas.

Fig. 8 shows counted HOs, PPs, and calculated PPR metric for analyzed eNBs within a specific network coverage area. According to the results presented in Fig. 8, it can be seen that the number of HOs and PPs depends on the characteristics of the selected area, where urban areas have an extremely high number of HOs and PPs due to having more eNBs and UEs than suburban and especially rural areas. As a consequence, larger PPR metrics are obtained for rural areas having a lower number of eNBs in comparison to urban areas. Besides the influence of UEs density on PPs and PPR metric, eNBs configuration has a large impact on a number of PPs.

Fig. 9 shows the number of UEs per each of the analyzed UE velocity intervals. It can be noticed that the highest number of UEs in urban areas are grouped in lower UE velocity intervals [20–50], while the highest number of UEs in suburban and especially rural areas are grouped in UE velocity intervals with somewhat higher values [60–100]. This is due to UEs moving across high-velocity roads as opposed to those in urban areas, which in turn contributes to such UEs velocity distribution.

5.2. The 3GPP ping-pong rate analysis

As defined by the 3GPP in [9], the PPR metric for eNB b , can be calculated as presented in Eq. (15).

$$PPR^b = \frac{|PP^b|}{|HO^b|} : b \in \mathbb{B} \quad (15)$$

Since eNBs can be configured to prefer certain UE velocities, deeper insights related to the occurrence of HOs, PPs, and the levels of PPR metric per UE velocity intervals are shown in Fig. 10. Fig. 10 presents

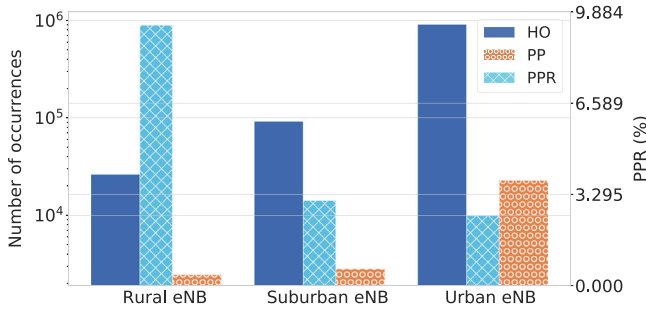


Fig. 8. HO, PP, and PPR values for selected eNBs.

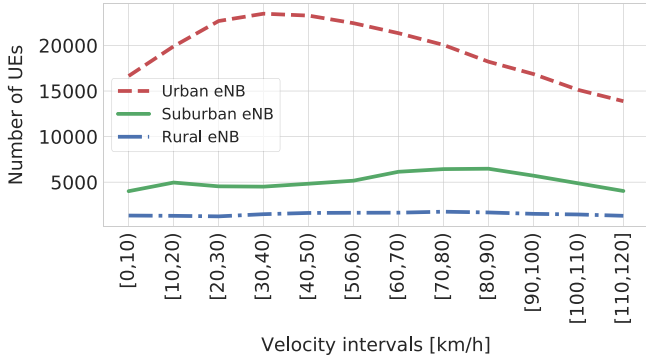


Fig. 9. Number of UEs per velocity intervals for selected eNBs.

results for urban, suburban, and rural areas. It can be seen that according to the current 3GPP definition, the PPR metric for the eNB located in the suburban area indicates better performance for higher UE velocities as opposed to lower ones, while urban and rural eNBs prefer lower UE velocities. For the eNB located in the rural area, HOs dominate for mid-range UE velocities, while the urban eNB has an increased number of HOs for low UE velocities ($[0, 10)$). These results can be explained by population movements and different area/UEs characteristics. In urban areas, UEs with UE velocities below 10 km/h dominate, while suburban areas are characterized by middle UE velocity intervals. Rural areas are mostly characterized by UE velocities above 30 km/h, with a slight decrease for UE velocities above 90 km/h.

To extend the previous analysis with the eNB coverage metric, HOs, PPs, and PPR values for urban, suburban, and rural eNB are shown in Fig. 11 in a two-dimensional format of heatmap expanded with bar charts on the secondary axis. More specifically, results in Fig. 11 are presented as a function of the UE velocity on the x-axis and the eNB coverage metric on the y-axis, where heatmap color intensity represents HOs, PPs, or PPR normalized values between 0 and 1 depending on what graph is observed. Consequently, the heatmap part of the graphs has the same color scale with different value-based intensities and results are easily comparable. Bar charts on the secondary x and y axis represent non-normalized (raw) HOs, PPs, and PPR values under different velocity and eNB coverage intervals, respectively. Moreover, outliers are removed in a way that all data where the number of HOs is under the 20th percentile of total HOs are removed from further analyses.

The results presented in Fig. 11 show that the highest HO peaks for eNBs in suburban and urban areas do not exist on PP and PPR graphs (values for the eNB coverage $[20, 30)$ and UE velocity $[10, 20)$ in suburban, and values for the eNB coverage $[70, 80)$ and UE velocity $[0, 10)$ in urban areas). This means that those HOs are unavoidable, i.e., they did not create the ping-pong effect, rather they simply occurred to fulfill the mobility requirement. Consequently, it can be noticed that a considerable portion of the occurred HOs is required to ensure basic 4G

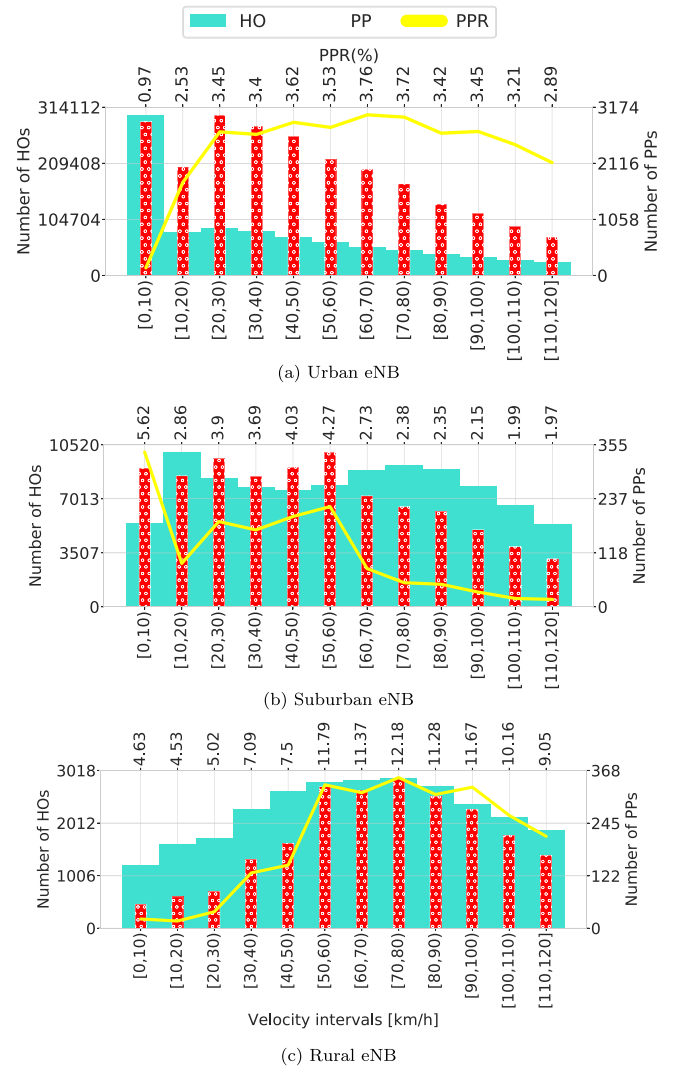


Fig. 10. Number of HOs, PPs, and PPR values per UE velocity intervals.

Table 3

Comparison of the 3GPP and novel ping-pong rate.

eNB	3GPP PPR (%)	Novel PPR (%)	Difference (%)
Urban	2.576	4.489	1.922
Suburban	3.116	5.255	2.139
Rural	9.413	10.511	1.098

Table 4

 R^2 values for fitting curves.

eNB	Logarithmic	Exponential	Linear
Urban	0.888	0.893	0.899
Suburban	0.955	0.956	0.921
Rural	0.875	0.889	0.856

network functionality. Furthermore, Fig. 11 shows that PPs occur where eNB coverage is low and that the probability of a HO being a PP is not the same for all HOs. Those non-equal probabilities and unavoidable HOs are the main shortcomings of the PPR metric defined by the 3GPP.

5.3. Novel comprehensive ping-pong rate metric

Novel $\widehat{PPR}$ metric is an empiric metric proposed in this paper, which mitigates the shortcomings of the PPR metric defined by the

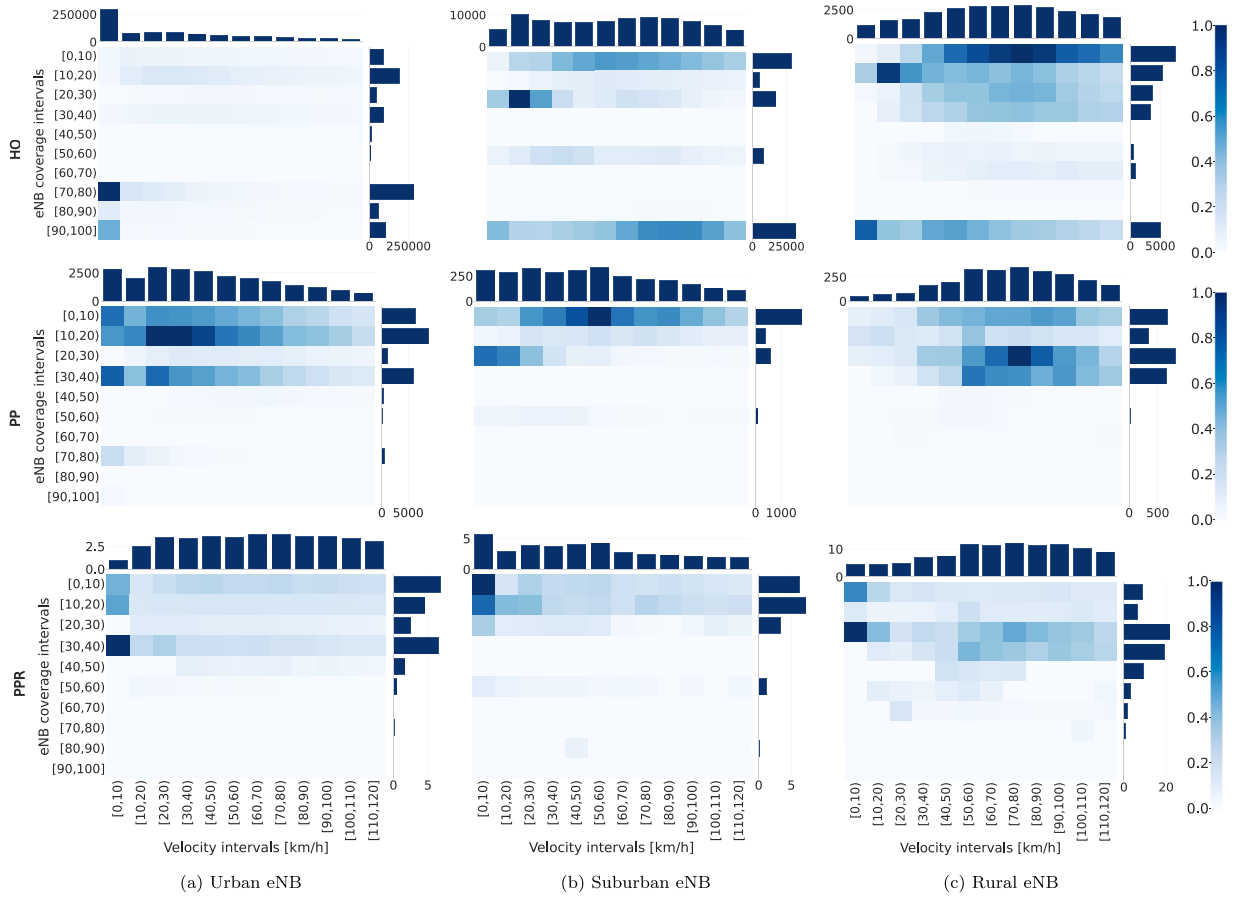


Fig. 11. Relation between normalized HO, PP, and PPR values per eNB coverage and UE velocity (heatmap). Relation between HO, PP, and PPR values per eNB coverage (y-axis, right hand side bar chart). Relation between HO, PP, and PPR values per UE velocity (x-axis, upper bar chart).

3GPP. Since the results from the previous section indicate that portions of HOs are unavoidable, they have a very low probability of being PPs. Additionally, the eNB coverage is highlighted as an important factor that reflects the probability that a HO may be a PP. Consequently, instead of calculating the share of PPs in all HOs, probability weights based on the eNB coverage for every HO are calculated.

All the below presented steps for calculating weights for eNB are presented in Algorithm 3. The first step in this process is calculating those weights by selecting an eNB of interest (*b*) and an area around it, which determines the subset of data used for the calculation. Since every eNB has a different coverage range [27], the subset data range is obtained by calculating a convex hull based on the locations of connected UEs (while removing outliers). The chosen area of interest is divided into multiple hexagons with corresponding data subsets and the selected eNB coverage is calculated based on Eq. (13) for every hexagon.

The second step includes the calculation of PPR values per eNB coverage using Eq. (16) such that:

$$PPR_d = \frac{|PP_d|}{|HO_d|} : d \in \mathbb{D} \quad (16)$$

where $|PP_d|$ and $|HO_d|$ represent the number of PPs and the number of HOs at eNB coverage d from $\mathbb{D}$. According to Eq. (16), the HOs and the PPs for eNB coverage are counted for the whole area and they are used to calculate the PPR metric. Fig. 12 depicts the relation between PPR metric and eNB coverage (by intervals). To extract the weights from this relation, linear, logarithmic, and exponential functions are used to fit PPR values per eNB coverage. The nonlinear least squares method is used for curve fitting, while R^2 (R squared) score is used for calculating the goodness of the fit [28]. A function that best fits PPR

per eNB coverage is defined in Expression (17) such that:

$$w : [0, 100] \rightarrow [0, 1] \quad (17)$$

w is best fit of (d, PPR_d) , $d \in \mathbb{D}$

where function w transforms the fitting probabilities percentage into the probability interval $[0, 1]$. It is interpreted as the probability $w(d)$ that a HO at the eNB coverage d might create the ping-pong effect for eNB b .

After finding the best fitting function (see Table 4), HOs and PPs are weighted based on their eNB coverage values and finally a novel $\widehat{PPR}$ metric for the selected eNB is calculated using Eq. (18) such that:

$$\widehat{PPR} = \frac{\sum_{d \in \mathbb{D}} (|PP_d| * w(d))}{\sum_{d \in \mathbb{D}} (|HO_d| * w(d))} \quad (18)$$

Results for fitting the PPR metric of selected areas are shown in Fig. 12. It is again noticeable that PPR decreases as eNB coverage decreases, with only a slight difference in the R^2 score for different areas and fitting functions. Furthermore, the linear function, compared to the logarithmic and the exponential one, is inapplicable, since it fits PPR into negative values as eNB coverage increases. This is not feasible since PPR is a ratio and its possible values range from 0 to 100.

Table 3 shows a comparison between the original 3GPP and the novel ping-pong rate. As seen from the table, the novel ping-pong rate ($\widehat{PPR}$) compared to 3GPP PPR is generally higher as a result of the decreased impact of unavoidable HOs.

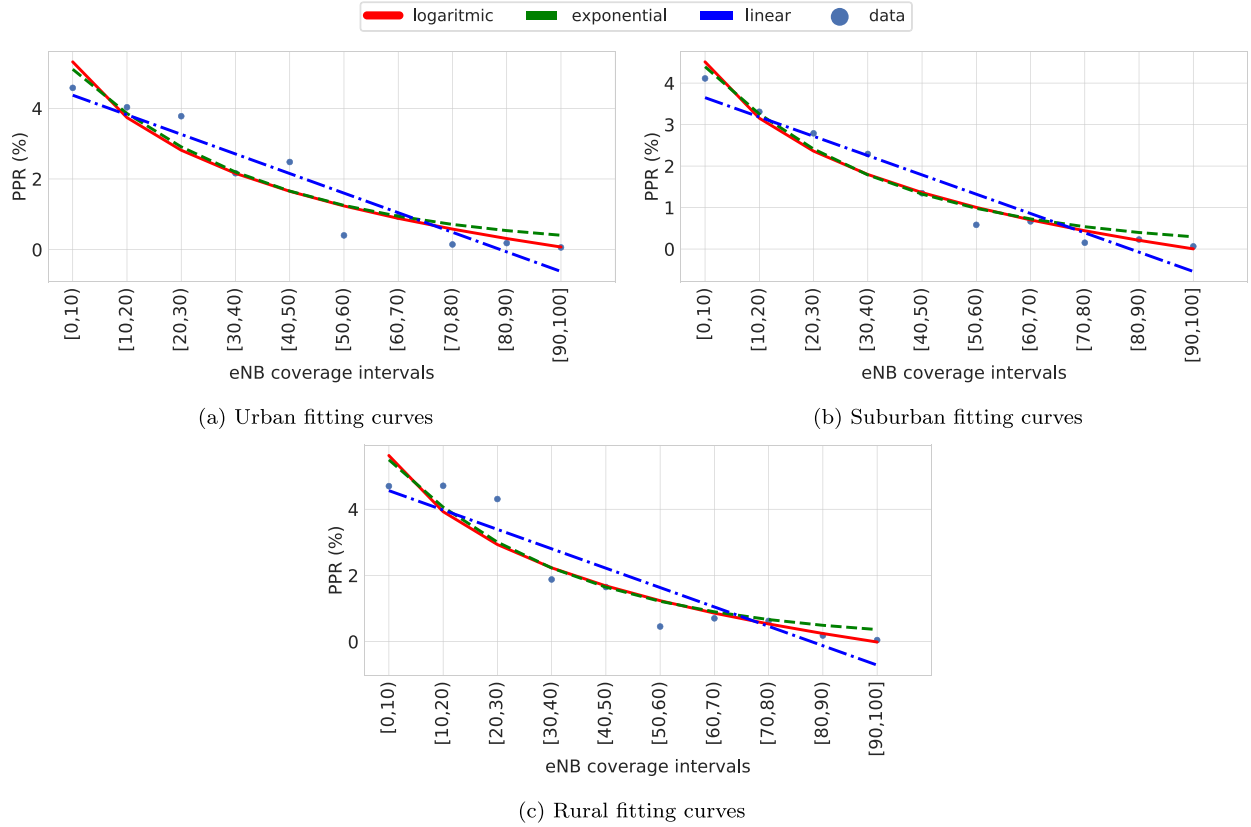
Fig. 13 shows the comparison of 3GPP PPR and the novel $\widehat{PPR}$ by UE velocity intervals. Since the impact of unavoidable HOs in $\widehat{PPR}$ is reduced, values per UE velocity intervals are increased. For Urban eNB, it can be seen that handling low-velocity UEs is not as good as it was initially indicated by PPR. Fig. 14 shows a comparison of PPR and

Algorithm 3: Calculating weights for eNB**Data:** All the records available for the area of interest**Result:** Probability for eNB that HO be PP per eNB coverage in the area of interest

```

1  $\forall a \in \mathbb{A}$                                      # for every area unit
2    $\forall b \in \mathbb{B}^{\mathbb{A}}$                              # for every eNB
3     Equation (13)                             # calculate eNB coverage
4    $\forall d \in \mathbb{D}$                                  # for every eNB coverage group
5     Algorithm 1 (d)                           # count HO that occur at eNB coverage d
6     Algorithm 2 (d)                           # count PPs that occur at eNB coverage d
7    $\forall d \in \mathbb{D}$                                  # for every eNB coverage group
8     Equation (16)                             # calculate PPR per eNB coverage
9   Expression (17)                             # assign weight per eNB coverage

```

**Fig. 12.** Fitting curves as a result of fitting PPR values per eNB coverage for selected areas.

$\widehat{PPR}$ by hours. It has a similar trend line, except for extremely larger values for late night hours. Only Rural eNB has almost the same values for PPR and $\widehat{PPR}$. Those values are the result of different population movements in different periods of the day, as well as the higher density of eNBs in urban and suburban areas.

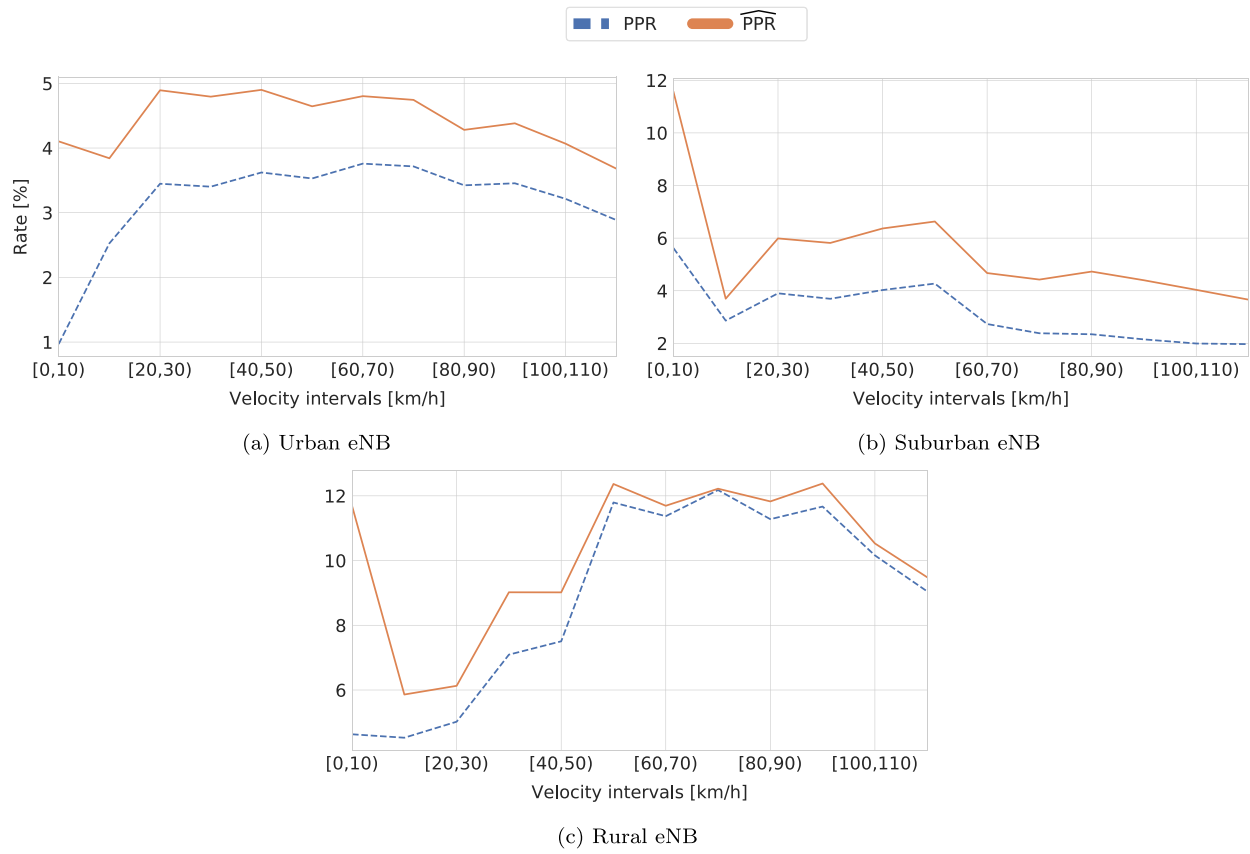
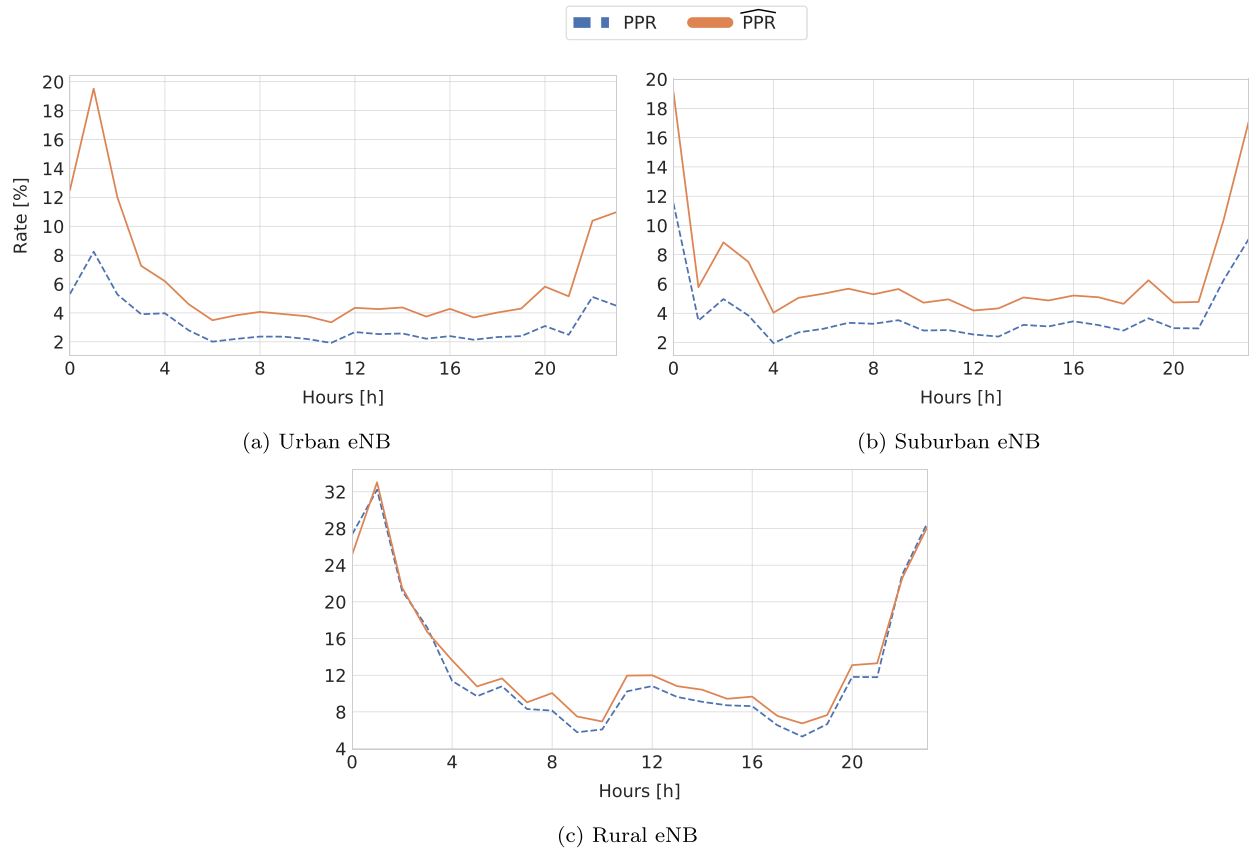
6. Discussion

Area density calculation performed to obtain an area type also produced some additional findings. First of all, it was successfully used to select an area of interest. Secondly, it indicates that highway area profiles are the same as the urban profile of large city, or even more urban than city areas, hence they are avoided in selection. Since highways connect cities, and people use them very often, they are areas with an increased flow of different UEs.

Since eNBs can be configured to prefer certain UE velocities, this points out that the UE velocity is an important factor during eNB configuration. However, in these analyses, eNB configuration is observed

as a black box. Using real data obtained from eNBs, the configurations are observed through PPs within predefined UE velocity intervals. The assumption that an eNB should be optimized for UE velocities of the majority of UEs seems invalid, since the majority of UEs with those UE velocities may usually be near the eNB where eNB coverage is high, hence HOs are not happening at all. This is especially important for UEs with low UE velocities that usually represent the majority of UEs in most eNBs since they move around the eNB very slowly and thus remain within the eNBs coverage area for a long time.

An increased number of HOs is not always the problem as well, since a HO is an essential process to ensure 4G mobility, i.e., it must occur from time to time. Consequently, every eNB has part of HOs that must occur, which are referred to as unavoidable HOs in this paper. If an increased number of HOs are mostly non-PPs, that indicates the unavoidable HOs. Consequently, the assumption is that an area with an increased number of non-PPs consists of multiple daily migrations from Point A to Point B, e.g., lots of people travel from a low density area (Point A) to a more densely populated area (Point B), where point A is

Fig. 13. PPR and $\widehat{PPR}$ comparison by UE velocity.Fig. 14. PPR and $\widehat{PPR}$ comparison by hours.

near one eNB, while point B is near another. Therefore, this results in a large number of HOs without creating PPs, i.e., HOs are unavoidable.

Furthermore, results in Section 5.2 also indicate that HOs and PPs occur mostly where eNB coverage is low, hence it is evident that along with the official 3GPP configuration parameters and UE velocity, the eNB coverage is an important factor for eNB optimization to ensure the low number of PPs. Additionally, this approach of including eNB coverage and UE velocity also provides an insight that PPR values must be taken with a certain amount of caution as they can lead to wrong conclusions. The reason for this is areas where a large number of PPs may seem insignificant simply due to an extremely high number of HOs that occur under high eNB coverage (i.e., they are unavoidable). In that case, PPR would indicate that an eNB is optimized for that certain UE velocity, while it may not be.

Accordingly, the most influential UEs in the context of the evaluation of an eNB configuration are UEs that are located in places where eNB coverage is low since they mostly result in high PPs and thus have a direct effect on the performance drop for all other UEs. Moreover, their UE velocities are also an important factor, since eNB can be configured to prefer certain UE velocities rather than others. Based on these findings, a novel *PPR* metric defined in Eq. (18) in Section 5.3 may be used as a guide to better optimize eNB configurations.

Finally, to improve our *PPR* metric, we may argue that if the connection signal strength (RSRP) information was available inside the dataset, the eNB coverage calculation would be improved. The calculation would provide more fine-grained results as every UE would have the exact measure of strength with its currently connected eNB, instead of only having a binary indicator whether it is connected or not. Consequently, strong RSRP would represent high eNB domination over a UE, while low RSRP would represent low eNB domination over that UE.

7. Conclusion

This paper presents the analyses that were performed to inspect eNB behavior concerning handovers, more specifically ping-pong handovers. The focus was given to identifying conditions in which ping-pong handovers mostly occur. The results show that it is not sufficient to consider only an eNB placement area, such as urban, suburban, rural, and highway, nor that the standard 3GPP metrics and handover configuration parameters are sufficient to create a well-configured eNB.

Further analysis shows that ping-pong handovers that affect eNB performance occur mostly on the edges of eNB coverage. Consequently, it represents a hotspot for reducing the number of PPs to ensure stable and reliable mobile networks with acceptable QoS for UEs. Based on these findings, a novel ping-pong rate was defined, where UE characteristics such as UE velocity and eNB coverage are used, with special emphasis on UEs on the edge of eNB coverage. The improved ping-pong rate provides a better insight as it reduces the impact of unavoidable handovers and thus provides a more realistic quality of the current eNB configuration.

For future work, listed findings may be used to investigate dynamic eNB configuration based on different times of the day. Additionally, population movement patterns can be explored to improve eNB configuration for better user satisfaction and improved resource allocations for telecom operators.

CRedit authorship contribution statement

Dinko Zidic: Methodology, Software, Validation, Formal analysis, Writing – original draft, Visualization, Investigation. **Toni Mastelic:** Conceptualization, Methodology, Writing – review & editing, Resources, Supervision, Project administration, Funding acquisition. **Ivana Nizetic Kosovic:** Methodology, Validation, Formal analysis, Writing – review & editing. **Mario Cagalj:** Methodology, Validation, Formal analysis, Writing – review & editing. **Josip Lorincz:** Methodology, Validation, Formal analysis, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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